JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATE, RTMNU & MSBTE Mumbai

Education to Eternity

Basic Science & Humanities Department

2023-24 (Semester)

MID SEM EXAMINATION

Course: B. Tech. (CSE/AI/DS)

Subject: NMAC

Subject Code:

Year/Semester: Ist Semester (First Year)

Max Marks: 20 **Date: - 26/ 10/2023** **Duration: - 1 Hr**

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

CO1. Describe concept of linear algebra, probability, logic, numerical method, distribution theory.

CO2. Illustrate the concept of linear algebra, probability, logic, numerical method, distribution theory by using examples.

CO3. Apply the knowledge of linear algebra, probability, logic, numerical method, distribution theory to solve the engineering problems.

CO4. Analyze the problems and results of linear algebra, probability, logic, numerical method, distribution theory to solve the real-world problems.

CO5. Evaluate the problems by linear algebra, probability, logic, numerical method, distribution theory.

CO6. Create the methods or model by linear algebra, probability, logic, numerical method, distribution theory.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Write the Simpson's 1/3 formula.	1/CO1	1
Q.2	Tell the formula of Trapezoidal rule.	1/CO1	1
Q.3	Write the Simpson's 3/8 formula.	1/CO1	1
Q.4	Find the number of sample points when 2 dies are thrown.	1/CO1	1

SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence)

Q.5	Solve to get the value of x_o , x_n and h from $\int_0^2 e^x \sin x \, dx$ with four strip.	3/CO3	2
Q.6	Write Axioms of probability.	2/CO2	2
Q.7	A random variable x as a density function given by	2/CO2	2

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$f(x) = \begin{cases} K\, e^{-3x} & x > 0 \\ 0 & x \leq 0 \end{cases}$ Find constant K.

Q.8	Define independent and dependent events.	1/CO1	2

SECTION C

1. Solve any two questions.

Q.9	Evaluate $\int_1^2 (4x+2)dx$ by Trapezoidal rule using two strips.	4/CO4	4
Q.10	Evaluate the corresponding value of x and y from the integral $\int_0^{1.5} \frac{x}{\sqrt{2+x^2}}\, dx$ for three strip.	3/CO3	4
Q.11	A factory manufacturing television has 4 units A, B, C and D. Then units manufacture 15%, 20%, 30% and 35% of the total output respectively .It was found that out of these out puts 1%, 2%, 2%, and 3% are defective. A TV is selected at random from the total output it was found to be defective, what is the probability that it was manufactured by unit C.	4/CO4	4

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